Introduction to Systematic Reviews and Meta-analysis

Outline

Introduction

What is a systematic review?

- A systematic identification and evaluation of all the available relevant evidence
- It attempts to:
  - identify all relevant studies fitting predefined criteria
  - systematically summarize the validity and findings of the studies
  - synthesize or integrate findings
- Governed by principles of **methodological rigor**, **transparency** and **reproducibility**

Steps in a systematic review

1. Formulate the research question
2. Decide eligibility criteria
3. Develop a protocol
4. Search for studies
5. Apply eligibility criteria to select relevant studies
6. Collect data from the studies
7. Assess studies for risk of bias/methodological quality
8. Summarize the studies
9. Synthesize findings of the studies (e.g. **meta-analysis**)
10. Interpret results and draw conclusion

Systematic Review of RCTs

- Randomized controlled trials (RCTs) are considered the most reliable source of evidence regarding relative treatment effects.
- However, different RCTs may give different (and often conflicting) findings to the same question
- These may be due to sampling error, differences in populations, definitions of intervention, etc.

What is a meta-analysis?

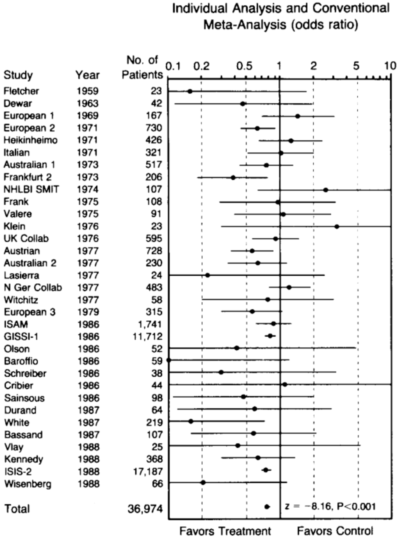
- It is a statistical method for combining the results from two or more studies with similar research question
- The idea is to implement a **weighted average** of the estimates according to the **precision** of each study
- We can use the **PICOT** framework to identify them:
  - **P** = Population
  - **I** = Intervention
  - **C** = Control/Comparison
  - **O** = Outcome
  - **T** = Time (follow-up)
- It is an *optional part* of a systematic review

Why do a meta-analysis?

- To quantify treatment effects and their uncertainty
- To increase power and precision
- To explore differences between studies

An example

This meta-analysis included 33 RCTs performed to assess the ability of streptokinase to prevent death following a heart attack.



Effect sizes

Some definitions

- **Effect size**: metric to quantify the difference of an outcome between two groups
- **Treatment effect**: effect size that quantifies the impact of a medical intervention between the group of treatment and the group of control
- *How to choose the effect size?* It should be:
  1. **comparable** between studies
  2. **computable** in studies
  3. with **good technical properties**, e.g. known sample distribution
  4. **interpretable**

Continuous outcome

Effect size: Standardized mean difference

If the outcome is measured differently across studies, there might be multiple scales of measurement. In such cases, we need to make the effect size comparable across studies:

$$\delta = \frac{\mu_1 - \mu_2}{\sigma}$$

$$d = \frac{\bar{X}_1 - \bar{X}_2}{S}$$

$$g = J \times d$$

$$V_g = J^2 \times V_d$$

where J is a correction factor ($0 < J < 1$)

Example: SMD

Let's see an example:

- $\bar{X}_1 = 103$, $\bar{X}_2 = 100$
- $n_1 = n_2 = 50$
- $S = 5.02$
- $J = 0.99$

Then:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{S} = \frac{103 - 100}{5.02} = 0.60; V_d = 0.04$$
$$g = J \times d = 0.99 \times 0.60 = 0.59$$
$$V_g = J^2 \times V_d = (0.99)^2 \times 0.04 = 0.04$$

Binary outcome

Effect size: Logarithm of the ratio between two odds

	Events	Non-Events	N
Treated	A	B	n_1
Control	C	D	n_2

where A = #events in treatment group, B = #non-events in treatment group, C = #events in control group, D = #non-events in control group

$$\text{OR} = \frac{\frac{A}{n_1 - A}}{\frac{C}{n_2 - C}} = \frac{\frac{A}{B}}{\frac{C}{D}} = \frac{AD}{BC}$$

$$V_{LogOR} = \frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D}$$

Example: Odds Ratio

Let's see an example

	Dead	Alive	N
Treated	5	95	100
Control	10	90	100

$$\text{OR} = \frac{AD}{BC} = \frac{5 \times 90}{95 \times 10} = 0.47$$

$$\log(\text{OR}) = \ln(0.47) = -0.76$$

$$V_{LogOR} = \frac{1}{A} + \frac{1}{B} + \frac{1}{C} + \frac{1}{D} = \frac{1}{5} + \frac{1}{95} + \frac{1}{10} + \frac{1}{90} = 0.32$$

Meta-analysis: fixed effects model

Meta-analysis: fixed effects model

- **Assumption**: the true effect size is the same across studies
- Observed treatment effects vary across studies due only to sampling error

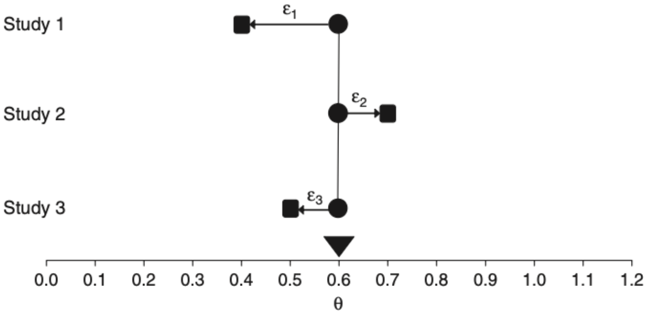


Figure: Fixed-effect model: observed vs true effect

Fixed effects model: Effect size

Defined θ the true effect and ϵ_i the sampling error from study i, the treatment effect can be written as:

$$Y_i = \theta + \epsilon_i$$

The meta-analyzed estimate of the effect size comes from the following weighted mean (inverse variance method):

$$M = \frac{\sum_{i=1}^{k} W_i Y_i}{\sum_{i=1}^{k} W_i}$$

where $W_i = \frac{1}{V_{Y_i}}$, with $i = 1, \ldots, k$. V_{Y_i} is the variance of the treatment effect Y_i for study i.

Fixed effects model: Variance and inference

The variance of the summary effect is estimated as the reciprocal of the sum of the weights:

$$V_M = \frac{1}{\sum_{i=1}^{k} W_i}$$

Then, 95% lower and upper limits of the confidence interval for the summary effect are:

$$LL_M = M - 1.96 \times \sqrt{V_M}$$

$$UL_M = M + 1.96 \times \sqrt{V_M}$$

Continuous outcome: example

	Treated			Control		
Study	Mean	SD	*n*	Mean	SD	*n*
Carroll	94	22	60	92	20	60
Grant	98	21	65	92	22	65
Peck	98	28	40	88	26	40
Donat	94	19	200	82	17	200
Stewart	98	21	50	88	22	45
Young	96	21	85	92	22	85

FE meta-analysis: an example (1)

Study	Effect size Y	Variance V_Y	Weight W	WY
Carroll	0.095	0.033	30.352	2.869
Grant	0.277	0.031	32.568	9.033
Peck	0.367	0.050	20.048	7.349
Donat	0.664	0.011	95.111	63.190
Stewart	0.462	0.043	23.439	10.824
Young	0.185	0.023	42.698	7.906
Sum			244.215	101.171

FE meta-analysis: an example (2)

$$M = \frac{101.171}{244.215} = 0.4143$$

$$V_M = \frac{1}{244.215} = 0.0041$$

$$SE_M = \sqrt{0.0041} = 0.0640$$

$$LL_M = 0.4143 - 1.96 \times 0.0640 = 0.2889$$

$$UL_M = 0.4143 + 1.96 \times 0.0640 = 0.5397$$

FE meta-anaysis: an example (3)

Hedges' *g* (Fixed effect)

	Std Diff (*g*)	Relative Weight
Carroll	0.10	12%
Grant	0.28	13%
Peck	0.37	8%
Donat	0.66	39%
Stewart	0.46	10%
Young	0.19	18%
Summary	0.41	100%

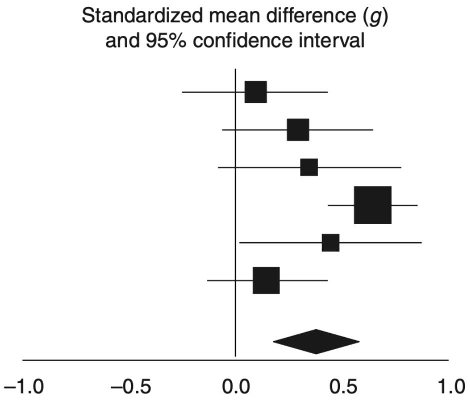


Figure: Forest plot – fixed-effect weights

Meta-analysis: random effects model

Within versus between study variance

- With (naturally) clustered data it is important to distinguish the variance *within* from the variance *between*
- Example of clustered data: patients from multiple studies or hospitals
- **Within-study variance** = variability inside the study, related mainly to sampling error (σ^2)
- **Between-study variance** = variability between studies, due to real differences between studies (τ^2)

Meta-analysis: random effects model

The true effect size varies from study to study, and the summary effect is the estimate of the distribution of effect sizes

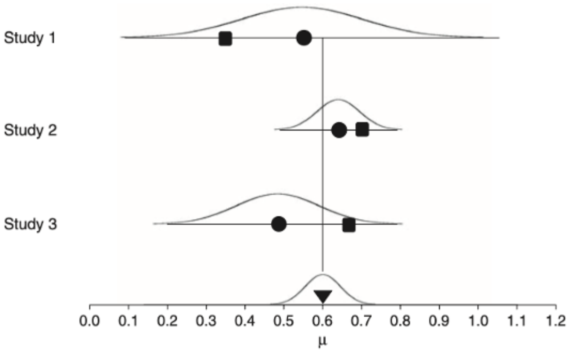


Figure: Random-effects model: between-study and within-study variance.

Random effects model: Effect size

In each study i, the observed effect Y_i can be written as $Y_i = \mu + \zeta_i + \epsilon_i$, where:

- μ is the grand mean
- ζ_i is the deviation of the study's true effect (between-study variance, heterogeneity)
- ϵ_i is the deviation of the study's observed effect from the study's true effect (within-study variance)

Each study is weighted by the inverse of its variance, that includes the sampling error V_{Y_i} and the heterogeneity τ^2:

$$V^*_{Y_i} = V_{Y_i} + \tau^2$$

$$W^*_i = \frac{1}{V^*_{Y_i}}$$

Random effects model: Pooled estimate

The weighted mean is the sum of the effect sizes multiplied by their weights divided by the sum of weights:

$$M^* = \frac{\sum_{i=1}^{k} W_i^* Y_i}{\sum_{i=1}^{k} W_i^*}$$

The variance of the summary effect is estimated as the reciprocal of the sum of the weights:

$$V_{M^*} = \frac{1}{\sum_{i=1}^{k} W_i^*}$$

$$SE_{M^*} = \sqrt{V_{M^*}}$$

Random effects model: inference and prediction

The 95% confidence interval for the summary effect is:

$$CI_{M^*} = M^* \pm 1.96 \times SE_{M^*}$$

The prediction interval is:

$$PI_{M^*} = M^* \pm t_{df}^{\alpha} \times \sqrt{T^2 + V_{M^*}}$$

where t_{df}^{α} is the α percentile of a t-student distribution with $k-2$ degrees of freedom.
The prediction interval is the interval within which a new study's effect would fall if this study was selected at random from this population

Random effects model: heterogeneity

- There are multiple ways to estimate the heterogeneity τ^2. The most popular are the Restricted Maximum Likelihood (REML), Paule-Mandel and DerSimonian-Laird
- How can we decide if there's heterogeneity or not? We can use the statistic I^2:

$$I^2 = \frac{\tau^2}{\tau^2 + V_Y} \times 100\%$$

- I^2 reflects the proportion of total variability explained by between-study variance. It ranges from 0% (all the variance is spurious) to 100% (there is heterogeneity)

RE meta-analysis: an example (1)

Let's see an example using the same data as before. We estimated τ^2 using DerSimonian-Laird estimator and obtained $T^2 = 0.037$.

Study	Y	V_Y	$V_Y + T^2$	W^*	W^*Y
Carroll	0.095	0.033	0.070	14.233	1.345
Grant	0.277	0.031	0.068	14.702	4.078
Peck	0.367	0.050	0.087	11.469	4.204
Donat	0.664	0.011	0.048	20.909	13.892
Stewart	0.462	0.043	0.080	12.504	5.774
Young	0.185	0.023	0.061	16.466	3.049
Sum				90.284	32.342

RE meta-analysis: an example (2)

$$M^* = \frac{32.342}{90.284} = 0.3582$$

$$V_{M^*} = \frac{1}{90.284} = 0.0111$$

$$SE_{M^*} = \sqrt{0.0111} = 0.1052$$

$$LL_{M^*} = 0.3582 - 1.96 \times 0.1052 = 0.1520$$

$$UL_{M^*} = 0.3582 + 1.96 \times 0.1052 = 0.5645$$

RE meta-analysis: an example (3)

Hedges' *g* (Random effects)

	Std Diff (*g*)	Relative Weight
Carroll	0.10	16%
Grant	0.28	16%
Peck	0.37	13%
Donat	0.66	23%
Stewart	0.46	14%
Young	0.19	18%
Summary	0.36	100%

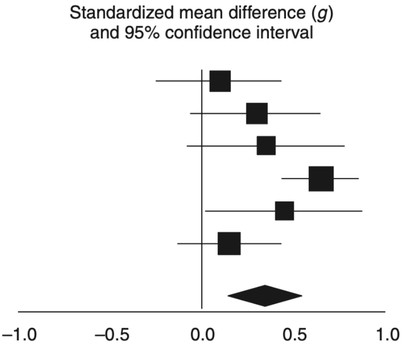


Figure: Forest plot – random-effects weights

Fixed vs Random effect

	Fixed effects	**Random effects**
True effect size	Same in each study	Varies across trials
Study contributions	Large studies contribute more, while small studies can be ignored	Small studies are not always ignored (between-study variance)
Error goes to 0 with	Infinite sample size	Infinite sample sizes and an infinite number of studies
To use if...	Studies are believed to be functionally identical and the goal is to evaluate the effect size for an identified population	Studies are believed to significantly differ and the goal is to generalize to a range of scenarios

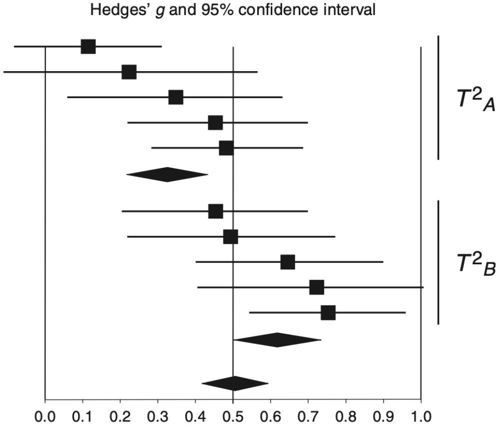
Subgroup meta-analysis

Subgroup analysis and meta-regression

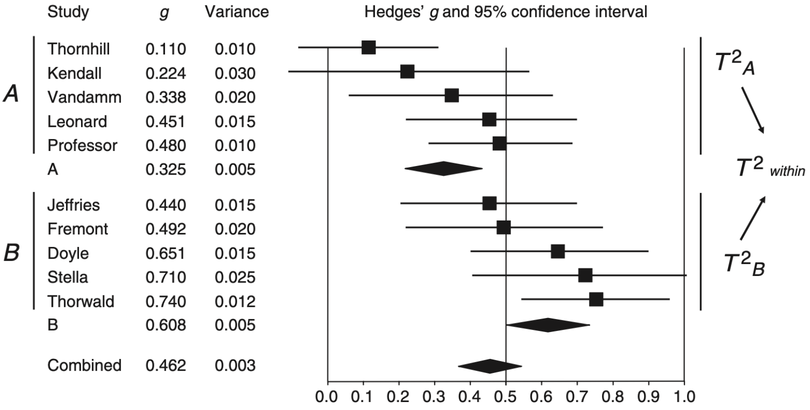
- The aim of subgroup analysis and meta-regression is to examine the association between study-level characteristics (covariates) and treatment effects
- Covariates must be decided with a scientific rationale
- The number of subgroups must be kept low to avoid issues related to multiple testing
- Two options are available:
  1. **Fixed effect meta-analysis**: we assume that all studies in subgroup A share a common effect and studies in subgroup B share a common effect
  2. **Random effect meta-analysis**: heterogeneity τ^2 must be estimated within subgroups, then it can be pooled or not when we perform a meta-analysis

Subgroup analysis: random effects with separate estimates of the between-study variance

	Study	g	Variance
A	Thornhill	0.110	0.010
	Kendall	0.224	0.030
	Vandamm	0.338	0.020
	Leonard	0.451	0.015
	Professor	0.480	0.010
	A	0.325	0.006
B	Jeffries	0.440	0.015
	Fremont	0.492	0.020
	Doyle	0.651	0.015
	Stella	0.710	0.025
	Thorwald	0.740	0.012
	B	0.610	0.004
	Combined	0.505	0.002



Subgroup analysis: random effects with pooled between-study variance

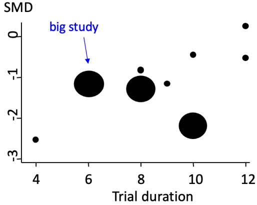


Meta-regression

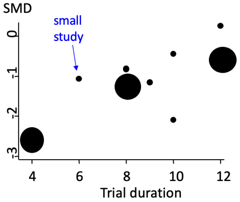
- Describes a linear relationship between **outcome variable** and **explanatory variable(s)**
- Assume study i reports y_i and has covariate x_i, a random effects meta-regression is written as:

$$y_i = \alpha + \beta x_i + \epsilon_i + \delta_i$$

  where ϵ_i is the random error and δ_i is the random effect
- Studies have different influences on the analysis and are weighted according to their precision



and



Summary and take home message

Take home message

- To perform a Systematic Review and a Meta-analysis it is important to start with a **clear research question** (PICOT framework)
- **Meta-analysis**: calculate a *summary* effect estimate which is a weighted average of the estimated treatment effects from individual studies
  - Weight studies according to the information they provide!
- **Fixed-effect meta-analysis**:
  - assumes treatment effect is the same in each study
  - inverse variance weights: $W_i = \frac{1}{V_{Y_i}}$
- **Random-effect meta-analysis**:
  - assumes treatment effect varies between studies
  - inverse variance weights: $W_i^* = \frac{1}{V_{Y_i} + \tau^2}$
- Suggested reading: Introduction to Meta-Analysis (Borenstein et al., 2009)